User's Manual

Model YN53
Pressure Transmitters

IM 01T01B04-01E

YOKOGAWA
Yokogawa Electric Corporation

IM 01T01B04-01E
3rd Edition

CONTENTS

1. INTRODUCTION 1-1

■ Regarding This Manual 1-1
1.1 For Safe Use of Product 1-1
1.2 Warranty 1-2

2. Checking the Instrument 2-1

2.1 Model and Specifications Check 2-1
2.1.1 Check the Items Received 2-1
2.1.2 Perform an External Visual Check 2-1
2.1.3 Check the Model and Suffix Codes 2-1
2.2 Storage 2-1

3. GENERAL SPECIFICATIONS 3-1

4. INSTALLATION 4-1

4.1 Selecting the Installation Location 4-1
4.2 Mounting the Transmitter 4-1

5. WIRING 5-1

6. BEFORE OPERATION 6-1

6.1 Zero Adjustment 6-1
6.2 Overpressure 6-1

7. MAINTENANCE 7-1

7.1 Precautions and Notes 7-1
7.2 Calibration 7-1
7.3 Zero and Span Adjustment 7-1
7.4 Amplifier Unit Replacement 7-2
7.5 Insulation Resistance and Dielectric Strength Test Procedures 7-3

INSTALLATION AND OPERATING PRECAUTIONS FOR JIS FLAMEPROOF EQUIPMENT EX-B03E

3rd Edition: Jan. 2003(YK)
All Rights Reserved, Copyright © 1990, Yokogawa Electric Corporation

Blank Page

1. INTRODUCTION

Thank you for purchasing the YN53 electronic pressure transmitter.

The YN53 Pressure Transmitters are precisely calibrated at the factory before shipment. To ensure correct and efficient use of the instrument, please read this manual thoroughly and fully understand how to operate the instrument before operating it.

■ Regarding This Manual

- This manual should be passed on to the end user.
- The contents of this manual are subject to change without prior notice.
- All rights reserved. No part of this manual may be reproduced in any form without Yokogawa’s written permission.
- Yokogawa makes no warranty of any kind with regard to this manual, including, but not limited to, implied warranty of merchantability and fitness for a particular purpose.
- If any question arises or errors are found, or if any information is missing from this manual, please inform the nearest Yokogawa sales office.
- The specifications covered by this manual are limited to those for the standard type under the specified model number break-down and do not cover custom-made instruments.
- Please note that changes in the specifications, construction, or component parts of the instrument may not immediately be reflected in this manual at the time of change, provided that postponement of revisions will not cause difficulty to the user from a functional or performance standpoint.
- The following safety symbol marks are used in this manual:

WARNING

Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION

Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

IMPORTANT

Indicates that operating the hardware or software in this manner may damage it or lead to system failure.

NOTE

Draws attention to information essential for understanding the operation and features.

1.1 For Safe Use of Product

For the protection and safety of the operator and the instrument or the system including the instrument, please be sure to follow the instructions on safety described in this manual when handling this instrument. In case the instrument is handled in contradiction to these instructions, Yokogawa does not guarantee safety. Please give your attention to the followings.

(a) Installation

- The instrument must be installed by an expert engineer or a skilled personnel. The procedures described about INSTALLATION are not permitted for operators.
- In case of high process temperature, care should be taken not to burn yourself because the surface of body and case reaches a high temperature.
- The instrument installed in the process is under pressure. Never loosen the process connector bolts to avoid the dangerous spouting of process fluid.
- During draining condensate from the pressure-detector section, take appropriate care to avoid contact with the skin, eyes or body, or inhalation of vapors, if the accumulated process fluid may be toxic or otherwise harmful.
- When removing the instrument from hazardous processes, avoid contact with the fluid and the interior of the meter.
- All installation shall comply with local installation requirement and local electrical code.

(b) Wiring

- The instrument must be installed by an expert engineer or a skilled personnel. The procedures described about WIRING are not permitted for operators.

- Please confirm that voltages between the power supply and the instrument before connecting the power cables and that the cables are not powered before connecting.

(c) Operation

- Wait 10 min. after power is turned off, before opening the covers.

(d) Maintenance

- Please do not carry out except being written to a maintenance descriptions. When these procedures are needed, please contact nearest YOKOGAWA office.
- Care should be taken to prevent the build up of drift, dust or other material on the display glass and name plate. In case of its maintenance, soft and dry cloth is used.

1.2 Warranty

- The warranty shall cover the period noted on the quotation presented to the purchaser at the time of purchase. Problems occurred during the warranty period shall basically be repaired free of charge.
- In case of problems, the customer should contact the Yokogawa representative from which the instrument was purchased, or the nearest Yokogawa office.
- If a problem arises with this instrument, please inform us of the nature of the problem and the circumstances under which it developed, including the model specification and serial number. Any diagrams, data and other information you can include in your communication will also be helpful.
- Responsible party for repair cost for the problems shall be determined by Yokogawa based on our investigation.
- The Purchaser shall bear the responsibility for repair costs, even during the warranty period, if the malfunction is due to:

  - Improper and/or inadequate maintenance by the purchaser.
  - Failure or damage due to improper handling, use or storage which is out of design conditions.
  - Use of the product in question in a location not conforming to the standards specified by Yokogawa, or due to improper maintenance of the installation location.
  - Failure or damage due to modification or repair by any party except Yokogawa or an approved representative of Yokogawa.
  - Malfunction or damage from improper relocation of the product in question after delivery.
  - Reason of force majeure such as fires, earthquakes, storms/floods, thunder/lightening, or other natural disasters, or disturbances, riots, warfare, or radioactive contamination.

1.3 JIS Flameproof Type

The model YN53 pressure transmitter with option code /JF3, which has obtained certification according to technical criteria for explosion-protected construction of electric machinery and equipment (Standards Notification No. 556 from the Japanese Ministry of Labor) conforming to IEC standards, is designed for hazardous areas where inflammable gases or vapors may be present. (This allows installation in Division 1 and 2 areas)

To preserve the safety of flameproof equipment requires great care during mounting, wiring, and piping. Safety requirements also place restrictions on maintenance and repair activities. Users absolutely must read “Installation and Operating Precautions for JIS Flameproof Equipment” at the end of this manual.

CAUTION

When the fill fluid near the sensor part moves from within, the instrument outputs a failure signal either high or low of the specific signal. In that case, generate the alarm to identify that the failure signal is output since the event may invalidate the flameproof approval. If the optional integral indicator is equipped, the indicator identifies the alarm on its display. Therefore, no other alarm generation is necessary.

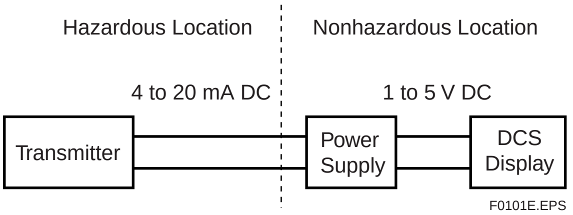


Figure 1.1 Example of using DCS (Distributed Control System)

2. Checking the Instrument

2.1 Model and Specifications Check

The YN53 pressure transmitters are thoroughly tested at the factory before shipment. When you receive this instrument, please check the following.

2.1.1 Check the Items Received

Check that the items in the following table are present.

Table 2.1 Item Check

Item	Qty.
Transmitter body	1
Bracket	1
U-bolt	1
U-bolt nuts	2
User's manual	1

T0201E.EPS

2.1.2 Perform an External Visual Check

When the transmitter is delivered, visually check them to make sure that no damage occurred during shipment.

2.1.3 Check the Model and Suffix Codes

The model name and specifications are indicated on the name plate attached to the case. If any problems or defects are found, please contact your nearest Yokogawa representative.

Table 2.2 Model and Suffix Codes

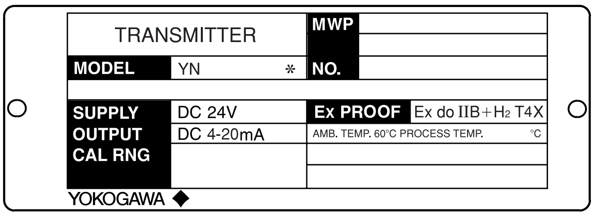
Model	Suffix Codes	Description
YN53	. . . .	General use type
YN53G	. . . .	Flameproof type
Output signal	-S	4 to 20 mA DC
Capsule (Span)	S.	5 to 20 kPa
	A.	50 to 200 kPa
	B.	130 to 500 kPa
	C.	0.25 to 1 MPa
	D.	0.5 to 2 MPa
	E.	1.3 to 5 MPa
	F.	2.5 to 10 MPa
	G.	5 to 20 MPa
	H.	13 to 50 MPa
Flange material	S.	SUS316 stainless steel
Process connection	1.	Rc1/4 female
	2.	Rc1/2 female
	3.	1/4NPT female
	4.	1/2NPT female
Style code	*A . .	Style A
Option code	/□	See Option codes

T0202E.EPS

Table 2.3 Optional Specifications

Item	Description	Code
Stainless steel bolt	SUS630 stainless steel mounting bolt for flange and bracket	SSB
Color change	Amplifier cover only See GS 22D01F01-00	SCF-□
Stainless steel tag plate	SUS304 stainless steel tag plate wired onto transmitter	TP-W
Oil-prohibited use	Degrease cleansing treatment	OSW
	Degrease cleansing treatment with fluorinated oil-filled capsule Operating temperature from −20°C	OSFC
JIS flameproof type	JIS flameproof type Ex do IIB+H2 T4X	JF3
Attached flameproof packing adapter	Electrical connection: G1/2 female Applicable cable O. D.: 8 to 12 mm	G11
Lightning protector	Transmitter power supply voltage: 12.5 to 32 V DC, Allowable voltage: 10 kV (1×40 μs) 50 times	LPT

T0203E.EPS



F0201E.EPS

Figure 2.1 Name Plate Example of JIS Flameproof Type

2.2 Storage

The following precautions must be observed when storing the instrument, especially for a long period.

(a) Select a storage area which meets the following conditions:
- It is not exposed to rain or water.
- It suffers minimum vibration and shock.
- It has an ambient temperature within the range of −40 to 100°C.

(b) When storing the transmitter, repack it as nearly as possible to the way it was packed when delivered from the factory.

(c) If storing a transmitter that has been used, thoroughly clean the chambers inside the process connector, so that no measured fluid remains in it. Also make sure before storing that the amplifier unit and capsule assembly are securely mounted.

3. GENERAL SPECIFICATIONS

Span and Range Limits:

Capsule	Measurement span	Measurement range	Maximum overpressure
S	5 to 20 kPa	0 to 20 kPa	100 kPa
A	50 to 200 kPa	−100 to 200 kPa	400 kPa
B	130 to 500 kPa	−100 to 500 kPa	1 MPa
C	0.25 to 1 MPa	−0.1 to 1 MPa	2 MPa
D	0.5 to 2 MPa	−0.1 to 2 MPa	4 MPa
E	1.3 to 5 MPa	−0.1 to 5 MPa	10 MPa
F	2.5 to 10 MPa	−0.1 to 10 MPa	20 MPa
G	5 to 20 MPa	−0.1 to 20 MPa	40 MPa
H	13 to 50 MPa	−0.1 to 50 MPa	75 MPa

T0301E.EPS

Output Signal:
Two wire 4 to 20 mA DC output

Accuracy:
A to H capsule: ±0.2% of Span
S capsule: ±0.5% of Span

Repeatability:
±0.02% of Span

Dead band:
±0.01% of Span

Temperature Limit:

Capsule		Ambient temp. (°C)	Process temp. (°C)
S	General use type	−20 to 80	−20 to 100
	Flameproof type	−20 to 60	
A to H	General use type	−40 to 100	−40 to 120
	Flameproof type	−20 to 60	−20 to 120

T0302.EPS

Ambient Humidity:
5 to 100%RH@40°C

Working Pressure Limits (Silicone Oil)
2.7 kPa abs to upper range limit for A to H capsule
0 kPa abs to upper range limit for S capsule

Supply Voltage :
12 to 42 V DC (See Figure 3.1)

Ambient Temperature Effect:
Total Effects per from 23°C to 50°C Change
±0.5% for maximum span (A to H capsule)
±1.25% for maximum span (S capsule)

Power Supply Effect:
±0.005%/V
(21.6 to 32 V DC, 350 Ω)

Mounting Position Effect:
0.17 kPa/90° (A to H capsule)
0.12 kPa/90° (S capsule)

Degrees of Protection:
IP65, NEMA 4, JIS C9020 Water tight protection

Exprosion Protected Type:
JIS flameproof type; Ex do IIB+H_2 T4X
Note: In case of the ambient temperature exceeds 55°C, use the heat-resistance cables with maximum allowable temperature of 70°C or above.

Mounting:
2-inch pipe mounting

Process Connections:
Rc 1/4, Rc 1/2, 1/4 NPT, or 1/2 NPT female

Electrical Connections:
G 1/2 or 1/2 NPT female

Bolt Material:
SCM435

Wetted Parts Material:
Diaphragm; SUS316L
Flange; SUS316
Capsule gasket (O-ring); Fluorinated rubber

Fill Fluid:
Silicone oil

Amplifier Housing:
Cast aluminium alloy
Polyurethane resin baked finish
Yellow Green (Munsell 5.0 GY3.6/1.3)

Zero Adjustment Limit:
A to H capsule;
Zero can be fully elevated or suppressed as long as low and high range value are within the measurement range limits of the capsule.
S capsule;
Zero can be fully elevated or suppressed as long as low and high range value are within the measurement range limits of the capsule and less than 200% of span.

Weight:
A to H capsule;
2.4 kg without 2-inch pipe bracket
S capsule;
3.0 kg without 2-inch pipe bracket

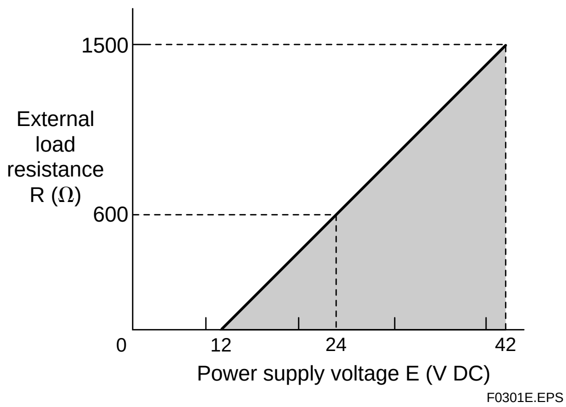


Figure 3.1 Relationship between Power Supply Voltage and External Load Resistance

4. INSTALLATION

4.1 Selecting the Installation Location

The transmitter is designed to withstand severe environmental conditions. However, to ensure stable and accurate operation for years, observe the following precautions when selecting an installation location.

(a) Ambient Temperature
Avoid locations subject to wide temperature variations or a significant temperature gradient. If the location is exposed to radiant heat from plant equipments, provide adequate thermal insulation and/or ventilation.

(b) Ambient Atmosphere
Avoid installing the transmitter in a corrosive atmosphere. If the transmitter must be installed in a corrosive atmosphere, there must be adequate ventilation as well as measures to prevent intrusion or stagnation of rain water in conduits.

(c) Shock and Vibration
Select an installation site suffering minimum shock and vibration (although the transmitter is designed to be relatively resistant to shock and vibration).

(d) Installation of Explosion-protected Transmitters
Explosion-protected transmitters can be installed in hazardous areas according to the types of gases for which they are certified. See Subsection 1.3 “JIS Flameproof Type.”

IMPORTANT

For the installation location of the transmitter, please read the “Installation and Operating Precautions for JIS Flameproof Equipment” at the end of this manual.

4.2 Mounting the Transmitter

See Figure 4.1 and 4.2.

IMPORTANT

Tighten the process connection bolts uniformly to a torque of 47 N·m {4.8 kgf·m} for A to H capsules, or a torque of 9.8 N·m {1 kgf·m} for S capsule.

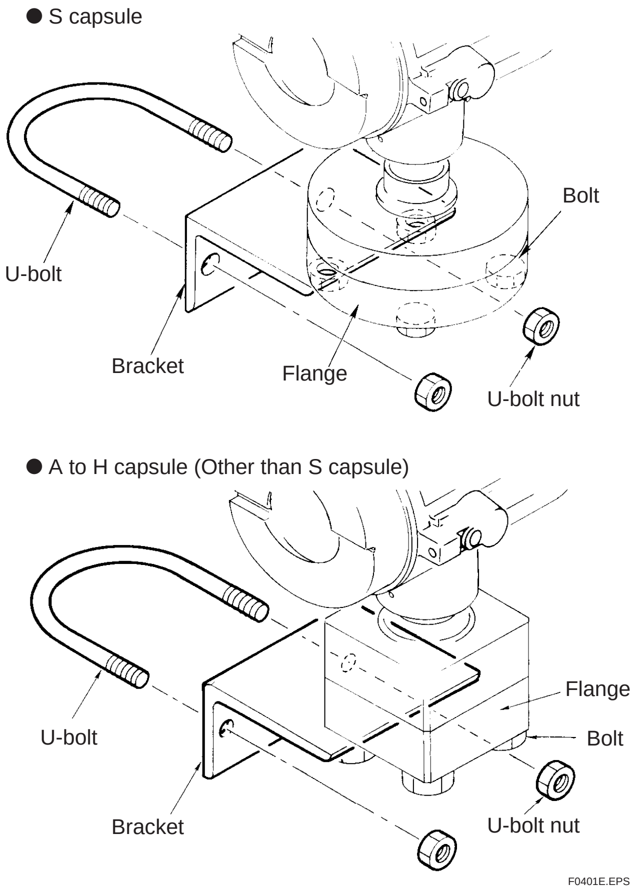


Figure 4.1 Transmitter Mounting Bracket

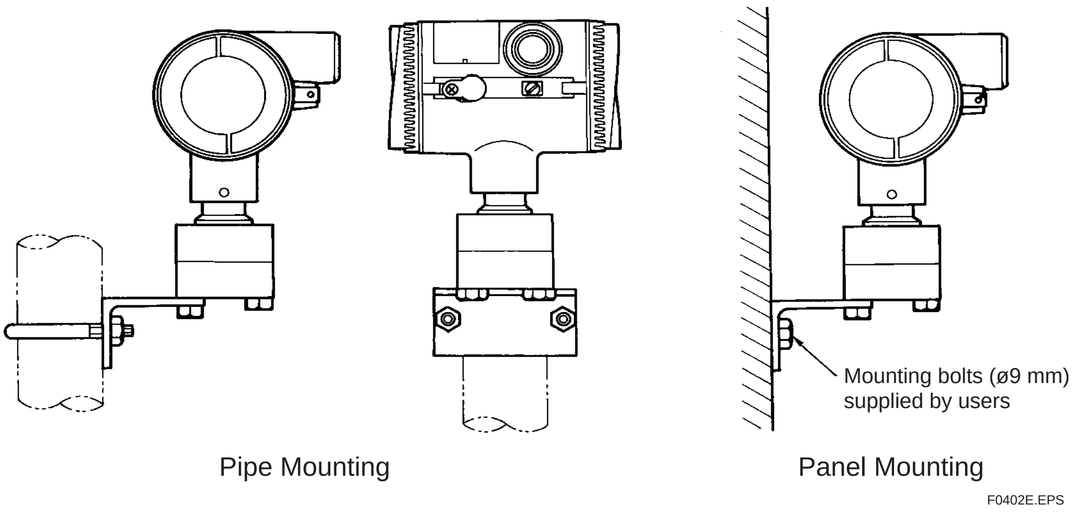


Figure 4.2 Transmitter Mounting

5. WIRING

The transmitter requires a DC power source. Connect the wiring as shown in Figure 5.1.

CAUTION

For the wiring materials and wiring work for the transmitter, please read the “Installation and Operating Precautions for JIS Flameproof Equipment” at the end of this manual.

■ Wiring Precautions

1) Avoid routing wiring near noise sources such as high-capacity transformers, motors, or motor power supplies.
2) For power wiring termination, crimp terminal lugs with insulating sleeves (for 4 mm screws) are recommended.
3) So that the rainwater, etc. cannot enter the instrument through the electrical connection port, use a watertight gland, thick electrical cable, flexible conduit and other such means when wiring.
4) The maximum load resistance between the transmitter and the distributor should be 600 Ω or less (for a 24 V DC power source).
5) Grounding is always required for the proper operation of transmitters. Follow the domestic electrical requirement as regulated by each country.

WARNING

For JIS flameproof type, grounding should satisfy Class D requirements (grounding resistance, Ω or less).

6) There are ground terminals on both the inside and outside of the case. Either of there may be used.
7) Shielded cable is recommended for wiring.

CAUTION

If the transmitter is flameproof and the ambient temperature is 55°C or more, use cables having a maximum allowable heat resistance of at least 70°C in consideration of the instrument’s generation of heat or the cables’ self-heating.

■ Mounting flameproof packing adapter body to conduit connection (see Figure 5.2)

1) Screw the flameproof packing adapter into the terminal box until the O-ring touches the wiring port (at least 6 full turns), and firmly tighten the lock nut.
2) Insert the cable through the union cover, the union coupling, the clamp nut, the clamp ring, the gland, the washer, the rubber packing, and the packing box, in that order.
3) Insert the end of the cable into the terminal box.
4) Tighten the union cover to grip the cable. When tightening the union cover, tighten approximately one turn past the point where the cable will no longer move up and down.
   Proper tightening is important. If it is too tight, a circuit break in the cable may occur; if not tight enough, the flameproof effectiveness will be compromised.
5) Fasten the cable by tightening the clamp nut.
6) Tighten the lock nut on the union cover.
7) Connect the cable wires to each terminal.

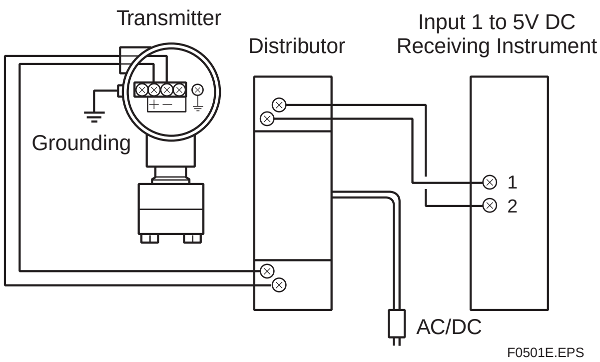


Figure 5.1 Connection between Transmitter and Distributor

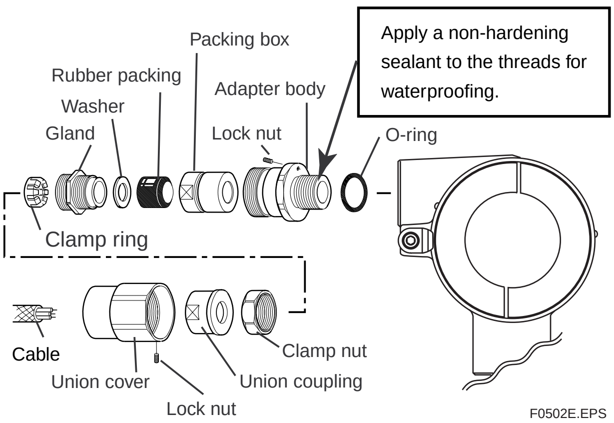


Figure 5.2 Installing Flameproof Packing Adapter

6. BEFORE OPERATION

6.1 Zero Adjustment

■ Zero Adjustment

The zero adjustment must be performed before using the transmitter.

1) Connect to a test ammeter or digital multimeter.
2) Turn on the power of all instruments and allow at least five minutes for them to warm up.
3) If the pressure range is zero-based, leave the transmitter input pressure port open to the atmosphere. If it is not zero-based, apply a pressure corresponding to 0% output. Under these conditions, turn the zero adjustment screw so that the output is 0% (Figure 6.1).

6.2 Overpressure

Allowable maximum overpressure varies depending on the capsule you selected. Should the pressure exceed the values in Table 6.1, it is possible to break the sensor. Proceed with caution when applying the pressure.

⚠ **CAUTION**

For the vacuum application, the lower range value is −100 kPa {−1 kgf/cm^2} for A to H capsules (other than S capsule) and 0 kPa {0 kgf/cm^2} for S capsule.

Table 6.1 Maximum Overpressure

Capsule	Maximum Overpressure	
	(kgf/cm^2)	(Pa)
S	1	100 k
A	4	400 k
B	10	1 M
C	20	2 M
D	40	4 M
E	100	10 M
F	200	20 M
G	400	40 M
H	750	75 M

T0601E.EPS

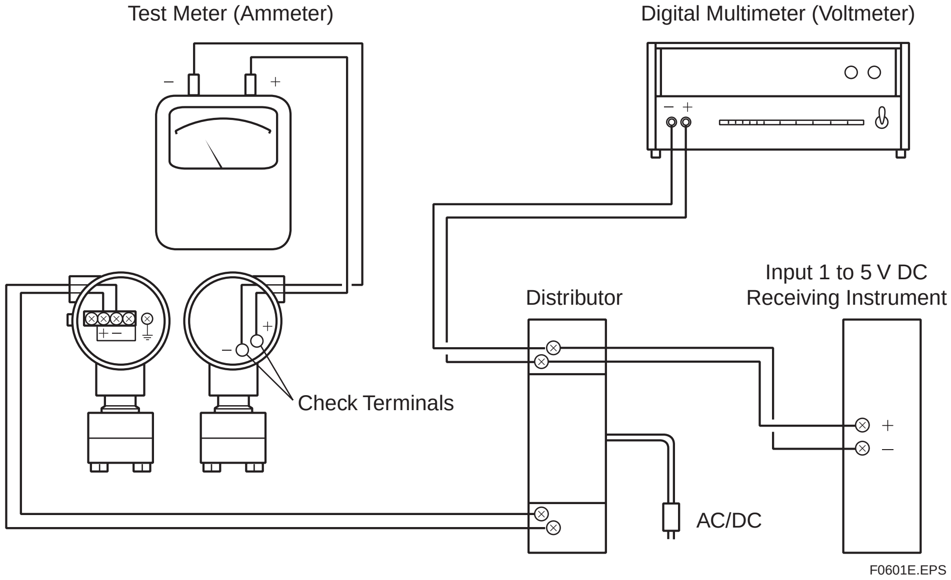


Figure 6.1 Instruments Connection

Table 6.2 Instruments Required for Calibration

Name	Yokogawa-recommended Instrument	Remarks
Power supply	Model SDBT or SDBS distributor	4 to 20 mA DC signal
Load resistor	Model 2792 standard resistor [250 Ω ±0.005%, 3 W]	
	Load adjustment resistor [100 Ω ±1%, 1 W]	
Voltmeter	Model 2501 A digital multimeter Accuracy (10 V DC range): ±(0.002% of rdg + 1 dgt)	
Digital manometer	Model MT220 precision digital manometer 1) For 10 kPa class Accuracy: ±(0.015% of rdg + 0.015% of F.S.) . . . . . for 0 to 10 kPa ±(0.2% of rdg + 0.1% of F.S.) . . . . for -10 to 0 kPa 2) For 130 kPa class Accuracy: ±0.02% of rdg . . . . for 25 to 130 kPa ±5digits . . . . for 0 to 25 kPa ±(0.2% of rdg + 0.1% of F.S.) . . . . for -80 to 0 kPa 3) For 700 kPa class Accuracy: ±(0.02% of rdg + 3digits) . . . . for 100 to 700 kPa ±5 digits . . . . for 0 to 100 kPa ±(0.2% of rdg + 0.1% of F.S.) . . . . for -80 to 0 kPa 4) For 3000 kPa class Accuracy: ±(0.02% of rdg + 10 digits) . . . . for 0 to 3000 kPa ±(0.2% of rdg + 0.1% of F.S.) . . . . for -80 to 0 kPa 5) For 130 kPa abs class Accuracy: ±(0.03% of rdg + 6 digits) . . . . for 0 to 130 kPa abs	Select a manometer having a pressure range close to that of the transmitter.
Pressure generator	Model 7674 pneumatic pressure standard for 200 kPa {2 kgf/cm^2}, 25 kPa {2500 mmH$_2$O} Accuracy: ±0.05% of F.S.	Requires air pressure supply.
	Dead weight gauge tester 25 kPa {2500mmH$_2$O} Accuracy: ±0.03% of setting	Select the one having a pressure range close to that of the transmitter.
Pressure source	Model 6919 pressure regulator (pressure pump) Pressure range: 0 to 133 kPa {1000 mmHg}	Prepare the vacuum pump for negative pressure ranges.

T0602E.EPS

7. MAINTENANCE

Maintenance of the transmitter is easy due to its modular construction. This chapter describes the procedures for calibration, adjustment, and the disassembly and reassembly procedures required for component replacement.

Since the transmitters are precision instruments, carefully and thoroughly read the following sections for proper handling during maintenance.

CAUTION

For the maintenance the transmitter, please read the “Installation and Operating Precautions for JIS Flameproof Equipment” at the end of this manual.

7.1 Precautions and Notes

When adjusting the instrument due to a change in the range, an amplifier unit replacement, or other such operation, these adjustments should as a rule be performed in a maintence room equipped with the necessary calibration equipment.

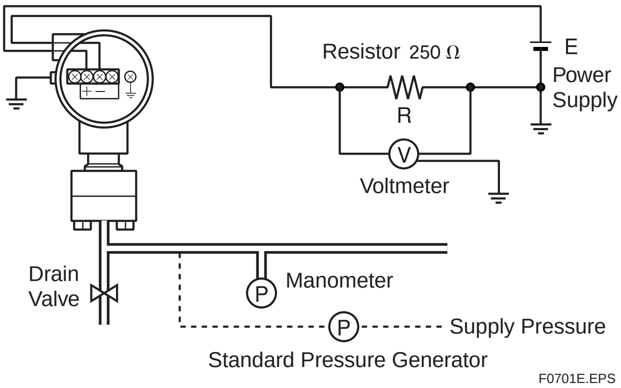


Figure 7.1 Connections for Maintenance

1) When adjusting for high accuracy, adjustments should be made with a test setup in which the load resistace, leadwires, power supply voltage, etc. are close to those conditions under which the transmitter will operate when actually installed.
2) Connect the wiring and piping as shown in Figure 7.1, and warm it up for at least five minutes.
3) Check that there is no leakage in the pressure supply piping.
4) Do not apply a pressure exceeding the equipment rating.

7.2 Calibration

Apply pressures corresponding to 0%, 25%, 50%, 75%, and 100% of the measurement range. When doing this, calculate the errors between the output obtained when the pressure is increased from 0% to 100%, and when it is decreased from 100% to 0%, and verify that these errors are within the accuracy specifications.

7.3 Zero and Span Adjustment

1) Refferring to Figure 7.1, connect the wiring and piping, and warm it up for at least five minutes.
2) Place the span coarse adjustment jumper in position at H (HIGH), M (MID), or L (LOW). Refer to the following table when selecting the jumper position.

Table 7.1 Selection of the Span Adjustment Jumper

Span Coarse Adjustment Jumper	Minimum Span	Maximum Span
H (HIGH)	0.61× Capsule Max. Span	Capsule Max. Span
M (MID)	0.35× Capsule Max. Span	0.35× Capsule Max. Span
L (LOW)	Capsule Max. Span	0.35× Capsule Max. Span

T0701E.EPS

Note: If the optimum span adjustment cannot be achieved at the selected jumper position, select another one.

3) To perform the zero adjustment, apply a pressure corresponding to 0% of the measurement range, and turn the zero adjustment screw so that the output is 4 mA (See Figure 7.2).
4) To perform the span adjustment, apply a pressure corresponding to 100% of the measurement range, and turn the span adjustment screw so that the output is 20 mA (See Figure 7.2).
5) Repeat 3) and 4) until the errors are within the accuracy specifications at both pressures.
6) Apply pressures corresponding to 0%, 25%, 50%, 75%, and 100% of the measurement range. When doing this, calculate the errors between the output obtained when it is decreased from 100% to 0%, and verify that these errors are within the accuracy specifications.

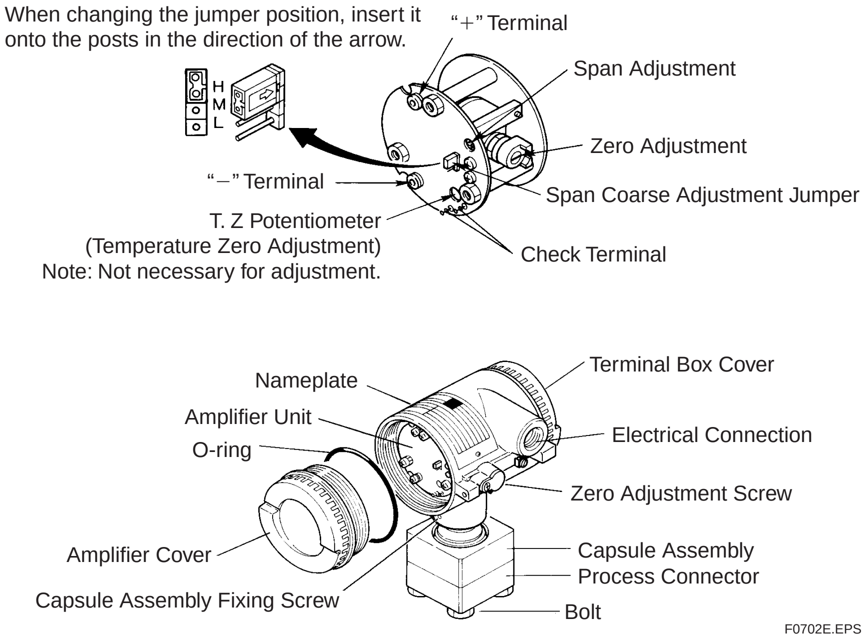


Figure 7.2 Component Names

7.4 Amplifier Unit Replacement

1) Turn the power off, and remove the amplifier cover.
2) Disconnect the "+" and "−" leadwires (See Figure 7.3).
3) Use a flat-blade screwdriver to turn the zero adjustment until the zero adjustment screw bracket is in the position shown in Figure 7.3.
4) Loosen the three amplifier unit mounting screws using a Phillips screwdriver.
5) Pull the amplifier unit straight out. Next, remove the connector.
6) When remounting the amplifier unit, exercise great care in aligning the zero adjustment screw pin with the bracket position.
7) Tighten the three amplifier unit mounting screws.
8) Reattach and tighten the "+" and "−" leadwires. These wires are;
   Red leadwire: To the + terminal
   Black leadwire: To the − terminal
9) Refer to Section 7.3, and carry out the zero and span adjustments.
10) Replace the cover.

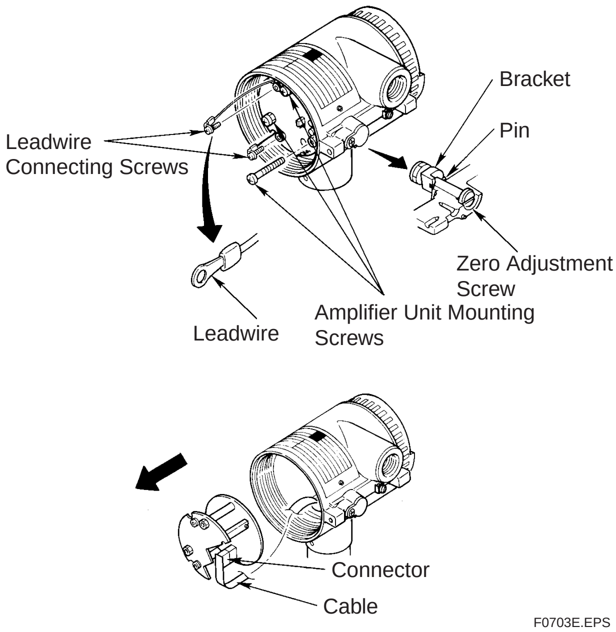


Figure 7.3 Removing Amplifier Unit

7.5 Insulation Resistance and Dielectric Strength Test Procedures

Connect a jumper wire between the SUPPLY "+" and "−" terminals, and apply the following rated voltages between those terminals and the ground terminal.

■ Insulation Resistance Test

At 500 V DC (1 minute or less), verify that the insulation resistance is 100 MΩ. After the test is completed, connect a 100 kΩ resistor between the jumper wire and ground terminal and discharge.

■ Dielectric Resistance Test

At 500 V DC (1 minute, set for 1 mA maximum), verify that there is no malfunction or abnormality.

NOTE

Performing the insulation resistance and dielectric strength tests too frequently may hasten deterioration of the electronic parts. It is recommended that these tests be performed only when the transmitter is not operating properly, and it is suspected that the cause of the problem lies in the electronic circuitry (the amplifier unit).

INSTALLATION AND OPERATING PRECAUTIONS FOR JIS FLAMEPROOF EQUIPMENT

Apparatus Certified Under Technical Criteria
(IEC-compatible Standards)

1. General

The following describes precautions on electrical apparatus of flameproof construction (hereinafter referred to as flameproof apparatus) in explosion-protected apparatus.

Following the Labour Safety and Health Laws of Japan, flameproof apparatus is subjected to type tests to meet either the technical criteria for explosionproof electrical machinery and equipment (standards notification no. 556 from the Japanese Ministry of Labour) (hereinafter referred to as technical criteria), in conformity with the IEC Standards, or the “Recommended Practice for Explosion-Protected Electrical Installations in General Industries,” published in 1979. These certified apparatus can be used in hazardous locations where explosive or inflammable gases or vapours may be present.

Certified apparatus includes a certification label and an equipment nameplate with the specifications necessary for explosion requirements as well as precautions on explosion protection. Please confirm these precautionary items and use them to meet specification requirements.

For electrical wiring and maintenance servicing, please refer to “Internal Wiring Rules” in the Electrical Installation Technical Standards as well as “USER’S GUIDELINES for Electrical Installations for Explosive Gas Atmospheres in General Industry,” published in 1994.

To meet flameproof requirements, equipment that can be termed “flameproof” must:

(1) Be certified by a Japanese public authority in accordance with the Labour Safety and Health Laws of Japan and have a certification label in an appropriate location on its case, and
(2) Be used in compliance with the specifications marked on its certification label, equipment nameplate and precautionary information furnished.

2. Electrical Apparatus of Flame-proof Type of Explosion-Protected Construction

Electrical apparatus which is of flameproof construction is subjected to a type test and certified by the Japanese Ministry of Labour aiming at preventing explosion caused by electrical apparatus in a factory or any location where inflammable gases or vapours may be present. The flameproof construction is of completely enclosed type and its enclosure shall endure explosive pressures in cases where explosive gases or vapours entering the enclosure cause explosion. In addition, the enclosure construction shall be such that flame caused by explosion does not ignite gases or vapours outside the enclosure.

In this manual, the word "flameproof" is applied to the flameproof equipment combined with the types of protection "e", "o", "i", and "d" as well as flameproof equipment.

3. Terminology

(1) Enclosure

An outer shell of an electrical apparatus, which encloses live parts and thus is needed to configure explosion-protected construction.

(2) Shroud

A component part which is so designed that the fastening of joint surfaces cannot be loosened unless a special tool is used.

(3) Enclosure internal volume

This is indicated by:— the total internal volume of the flameproof enclosure minus the volume of the internal components essential to equipment functions.

(4) Path length of joint surface

On a joint surface, the length of the shortest path through which flame flows from the inside to outside of the flameproof enclosure. This definition cannot be applied to threaded joints.

(5) Gaps between joint surfaces

The physical distance between two mating surfaces, or differences in diameters if the mating surfaces are cylindrical.

Note: The permissible sizes of gaps between joint surfaces, the path length of a joint surface and the number of joint threads are determined by such factors as the enclosure’s internal volume, joint and mating surface construction, and the explosion classification of the specified gases and vapours.

4. Installation of Flameproof Apparatus

(1) Installation Area

Flameproof apparatus may be installed, in accordance with applicable gases, in a hazardous area in Zone 1 or 2, where the specified gases are present. Those apparatus shall not be installed in a hazardous area in Zone 0.

Note: Hazardous areas are classified in zones based upon the frequency of the appearance and the duration of an explosive gas atmosphere as follows:
Zone 0: An area in which an explosive gas atmosphere is present continuously or is present for long periods.
Zone 1: An area in which an explosive gas atmosphere is likely to occur in normal operation.
Zone 2: An area in which an explosive gas atmosphere is not likely to occur in normal operation and if it does occur it will exist for a short period only.

(2) Environmental Conditions

The standard environmental condition for the installation of flameproof apparatus is limited to an ambient temperature range from –20°C to +40°C (for products certified under Technical Criteria). However, some field-mounted instruments may be certified at an ambient temperature up to +60°C as indicated on the instrument nameplates. If the flameproof apparatus are exposed to direct sunshine or radiant heat from plant facilities, appropriate thermal protection measures shall be taken.

5. External Wiring for Flameproof Apparatus

Flameproof apparatus require cable wiring or flameproof metal conduits for their electrical connections. For cable wiring, cable glands (cable entry devices for flameproof type) to wiring connections shall be attached. For metal conduits, attach sealing fittings as close to wiring connections as possible and completely seal the apparatus. All non-live metal parts such as the enclosure shall be securely grounded. For details, see the “USER’S GUIDELINES for Electrical Installations for Explosive Gas Atmospheres in General Industry,” published in 1994.

(1) Cable Wiring

- For cable wiring, cable glands (cable entry devices for flameproof type) specified or supplied with the apparatus shall be directly attached to the wiring connections to complete sealing of the apparatus.
- Screws that connect cable glands to the apparatus are those for G-type parallel pipe threads (JIS B 0202) with no sealing property. To protect the apparatus from corrosive gases or moisture, apply nonhardening sealant such as liquid gaskets to those threads for waterproofing.
- Specific cables shall be used as recommended by the “USER’S GUIDELINES for Electrical Installations for Explosive Gas Atmospheres in General Industry,” published in 1994.
- In necessary, appropriate protective pipes (conduit or flexible pipes), ducts or trays shall be used for preventing the cable run (outside the cable glands) from damage.
- To prevent explosive atmosphere from being propagated form Zone 1 or 2 hazardous location to any different location or non-hazardous location through the protective pipe or duct, apply sealing of the protective pipes in the vicinity of individual boundaries, or fill the ducts with sand appropriately.
- When branch connections of cables, or cable connections with insulated cables inside the conduit pipes are made, a flameproof or increased-safety connection box shall be used. In this case, flameproof or increased-safety cable glands meeting the type of connection box must be used for cable connections to the box.

(2) Flameproof Metal Conduit Wiring

- For the flameproof metal conduit wiring or insulated wires shall be used as recommended by the USER’S GUIDELINES for Electrical Installations for Explosive Gas Atmospheres in General Industry, published in 1994.
- For conduit pipes, heavy-gauge steel conduits conforming to JIS C 8305 Standard shall be used.
- Flameproof sealing fittings shall be used in the vicinity of the wiring connections, and those fittings shall be filled with sealing compounds to complete sealing of the apparatus. In addition, to prevent explosive gases, moisture, or flame caused by explosion form being propagated through the conduit, always provide sealing fittings to complete sealing of the conduit in the following locations:
  (a) In the boundaries between the hazardous and non-hazardous locations.
  (b) In the boundaries where there is a different classification of hazardous location.
- For the connections of the apparatus with a conduit pipe or its associated accessories, G-type parallel pipe threads (JIS B 0202) shall be used to provide a minimum of five-thread engagement to complete tightness. In addition, since these parallel threads do not have sealing property, nonhardening sealant such as liquid gaskets shall thus be applied to those threads for ensuring waterproofness.
- If metal conduits need flexibility, use flameproof flexible fittings.

6. Maintenance of Flameproof Apparatus

To maintain the flameproof apparatus, do the following. (For details, see Chapter 10 “MAINTENANCE OF EXPLOSION-PROTECTED ELECTRICAL INSTALLATION” in the USER’S GUIDELINES for Electrical Installations for Explosive Gas Atmospheres in General Industry.)

(1) Maintenance servicing with the power on.

Flameproof apparatus shall not be maintenance-serviced with its power turned on. However, in cases where maintenance servicing is to be conducted with the power turned on, with the equipment cover removed, always use a gas detector to check that there is no explosive gas in that location. If it cannot be checked whether an explosive gas is present or not, maintenance servicing shall be limited to the following two items:

(a) Visual inspection
Visually inspect the flameproof apparatus, metal conduits, and cables for damage or corrosion, and other mechanical and structural defects.

(b) Zero and span adjustments
These adjustments should be made only to the extent that they can be conducted from the outside without opening the equipment cover. In doing this, great care must be taken not to cause mechanical sparks with tools.

(2) Repair

If the flameproof apparatus requires repair, turn off the power and transport it to a safety (non-hazardous) location. Observe the following points before attempting to repair the apparatus.

(a) Make only such electrical and mechanical repairs as will restore the apparatus to its original condition. For the flameproof apparatus, the gaps and path lengths of joints and mating surfaces, and mechanical strength of enclosures are critical factors in explosion protection. Exercise great care not to damage the joints or shock the enclosure.

(b) If any damage occurs in threads, joints or mating surfaces, inspection windows, connections between the transmitter and terminal box, shrouds or clamps, or external wiring connections which are essential in flameproofness, contact Yokogawa Electric Corporation.

CAUTION

Do not attempt to re-process threaded connections or refinish joints or mating surfaces.

(c) Unless otherwise specified, the electrical circuitry and internal mechanisms may be repaired by component replacement, as this will not directly affect the requirements for flameproof apparatus (however, bear in mind that the apparatus must always be restored to its original condition). If you attempt to repair the flameproof apparatus, company-specified components shall be used.

(d) Before starting to service the apparatus, be sure to check all parts necessary for retaining the requirements for flameproof apparatus. For this, check that all screws, bolts, nuts, and threaded connections have properly been tightened.

(3) Prohibition of specification changes and modifications

Do not attempt to change specifications or make modifications involving addition of or changes in external wiring connections.

7. Selection of Cable Entry Devices for Flameproof Type

IMPORTANT

The cable glands (cable entry devices for flameproof type) conforming to IEC Standards are certified in combination with the flameproof apparatus. So, Yokogawa-specified cable entry devices for flameproof type shall be used to meet this demand.

References:

(1) Type Certificate Guide for Explosion-Protected Construction Electrical Machinery and Equipment (relating to Technical Standards Conforming to International Standards), issued by the Technical Institution of Industrial Safety, Japan

(2) USER’S GUIDELINES for Electrical Installations for Explosive Gas Atmospheres in General Industry (1994), issued by the Japanese Ministry of Labour, the Research Institute of Industrial Safet

Blank Page